EE6406: Analytic and Ensemble Learning (Lecture 2: Data Preprocessing)

Zhiping Lin

School of Electrical and Electronic Engineering
Nanyang Technological University

ezplin@ntu.edu.sg

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Lecture 2
Data Preprocessing

Overview

1. What is Data?
2. Data Types
3. Distance metrics
4. Preprocessing Methods

What is Data?

According to https://en.wikipedia.org/wiki/Data:

"In common usage, data are a collection of discrete or continuous values that convey information, describing the quantity, quality, fact, statistics, other basic units of meaning, or simply sequences of symbols that may be further interpreted formally. A datum is an individual value in a collection of data. "

Example 2.1

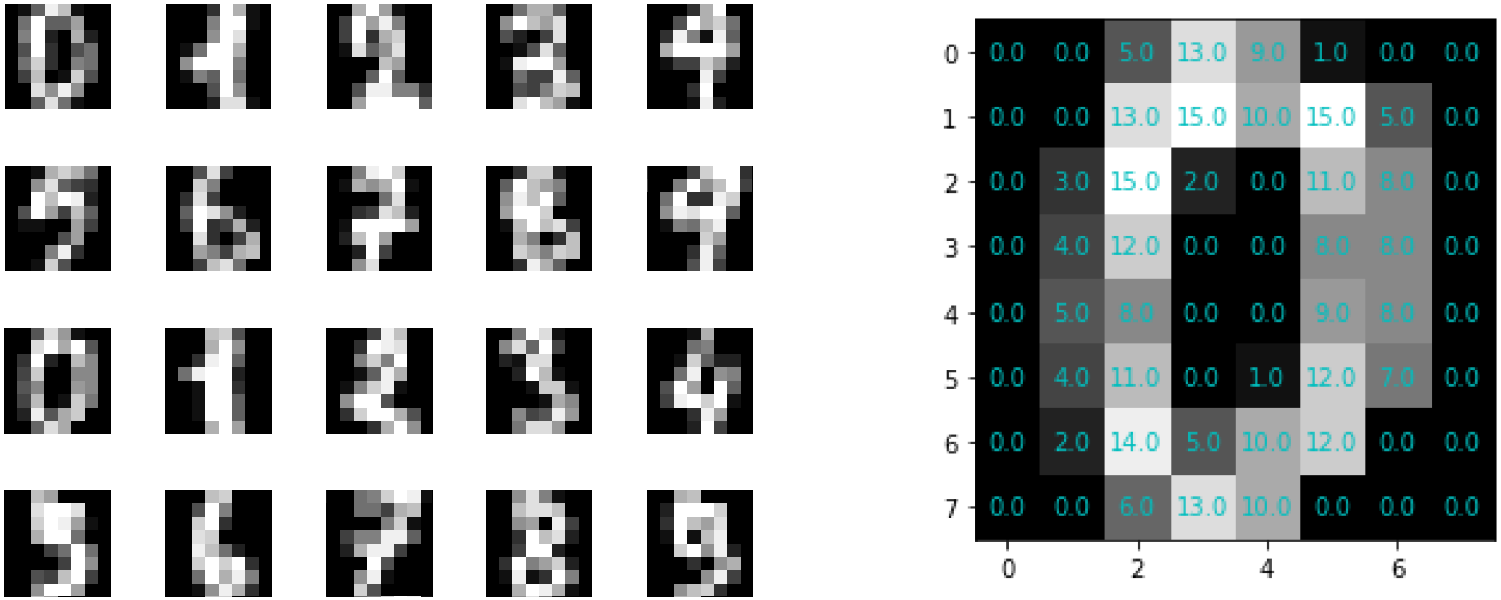


Figure 1: Left panel: samples of Optdigit dataset, right panel: the values in digit-0 sample.

Example 2.2

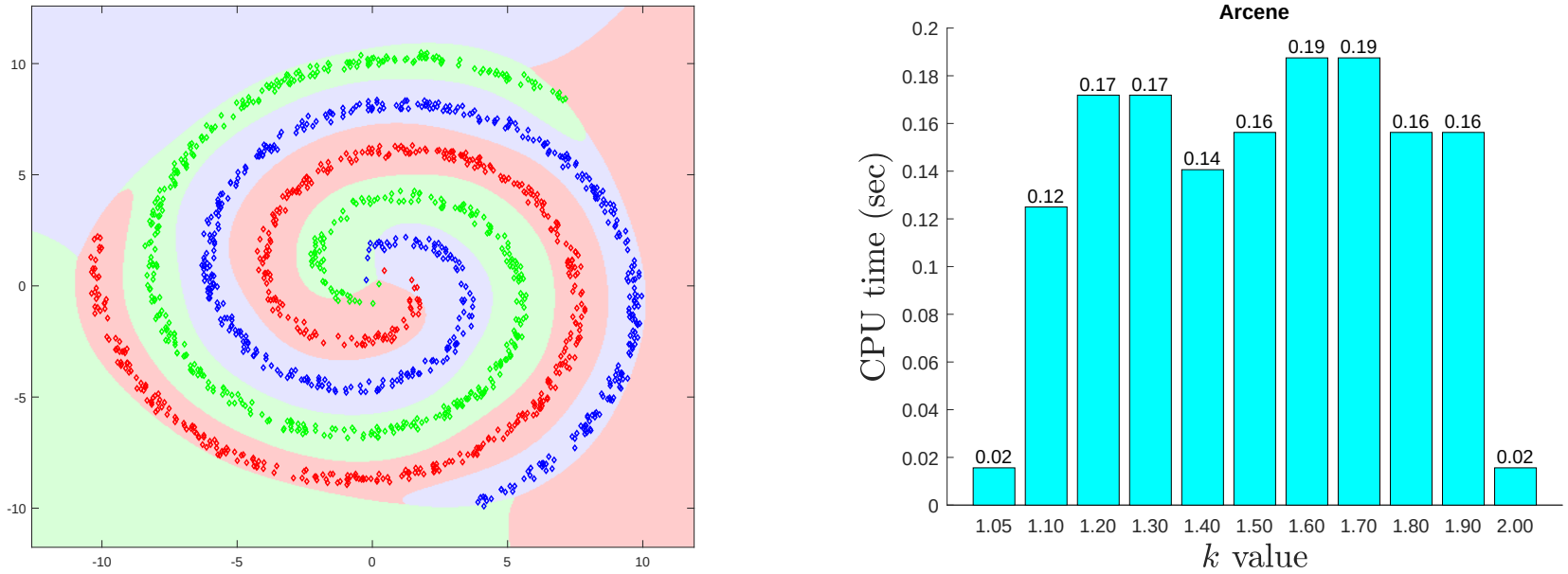
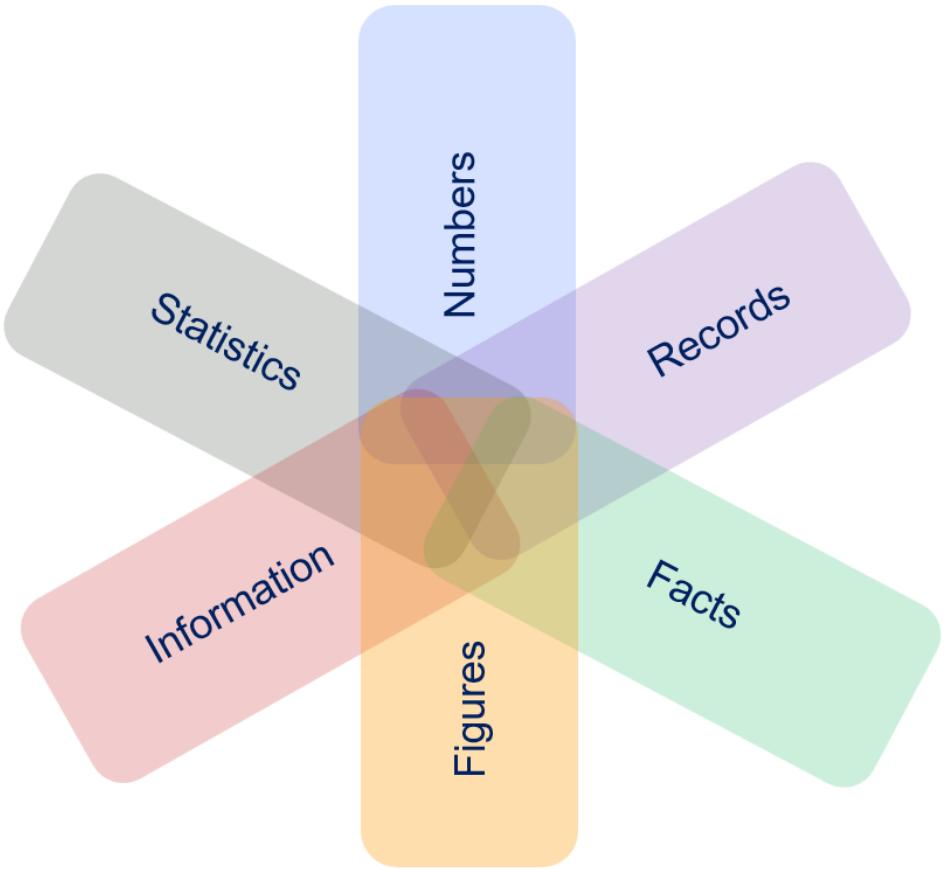


Figure 2: Left panel: a spiral pattern, right panel: a bargraph

What is data?

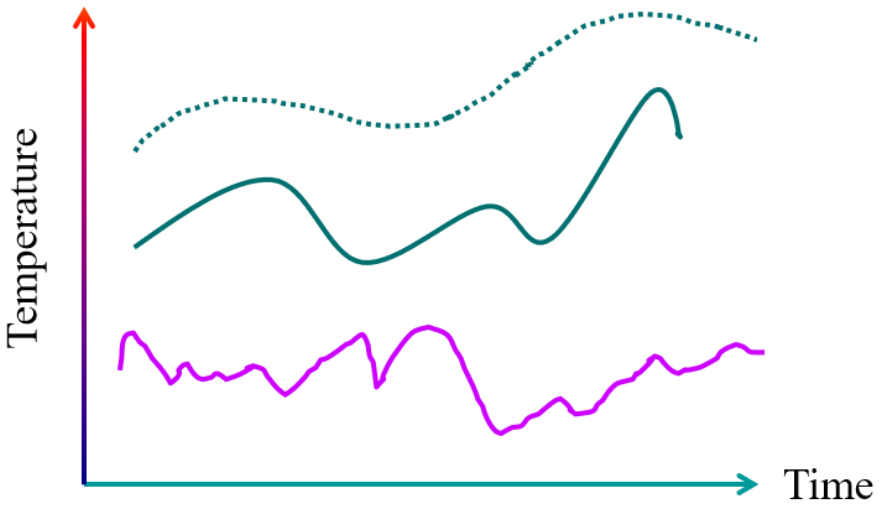
- **Continuous, Discrete**
- **Ordinal**
- **Categorical**
- **Missing**
- **Censored**



Example 2.3

Continuous, discrete variables

- Continuous variables are anything measured on a quantitative scale that could be any fractional number.
  - An example would be something like weight measured in kg.
- Discrete variables are numeric variables that have a countable number of values between any two values.



Example 2.4

Ordinal data

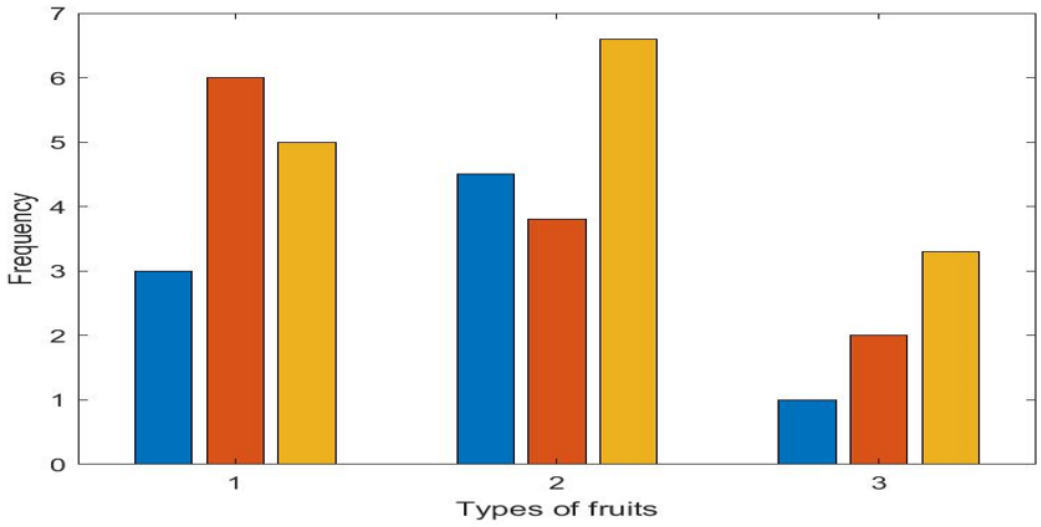
- Ordinal data are data that have a fixed, small number (< 100) of possible values, called levels, that are ordered (or ranked).
  - Example: survey responses

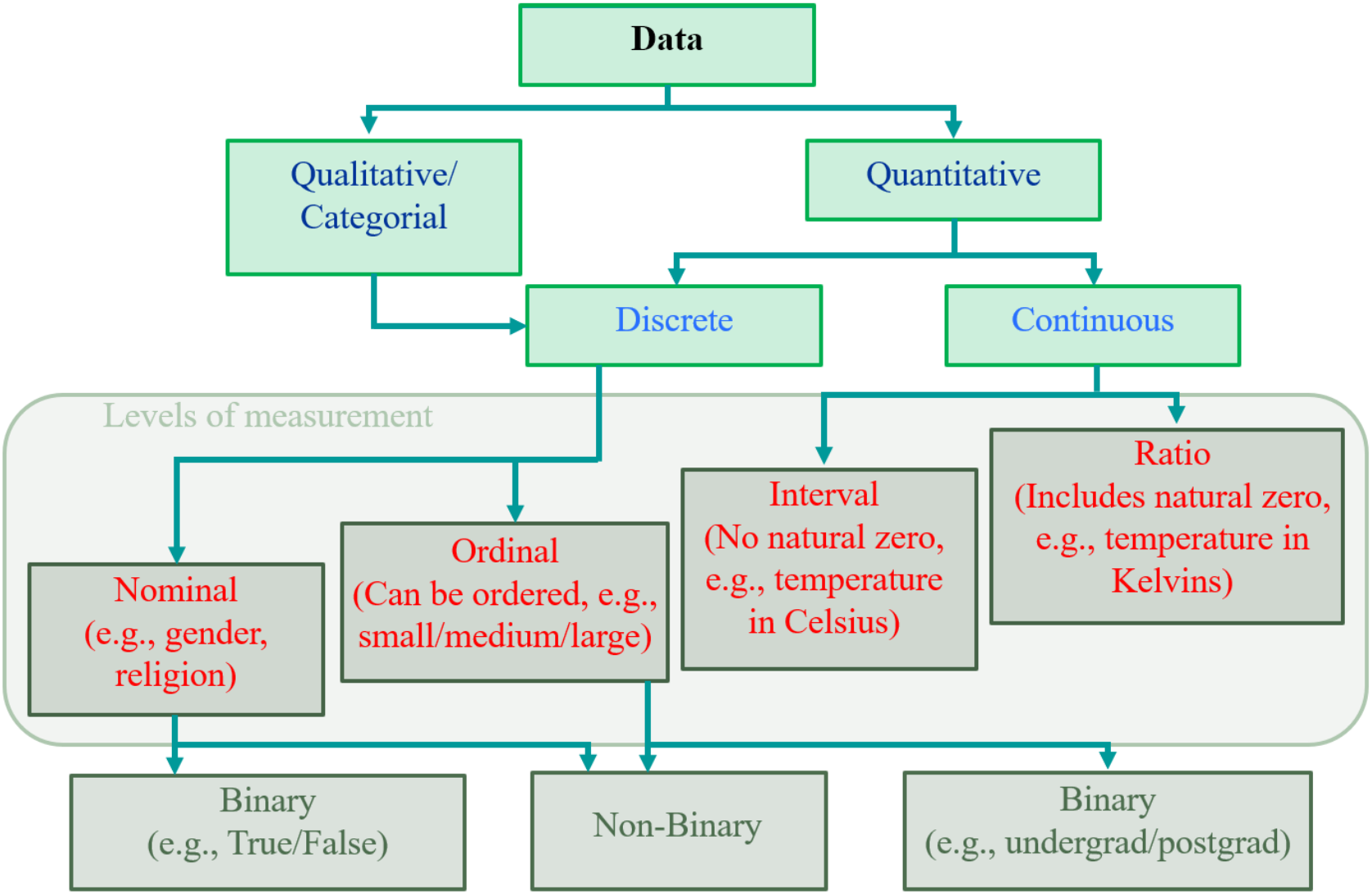
Excellent	Good	Average	Fair	Poor
✓				

Example 2.5

Categorical data

- Categorical data are data where there are multiple categories, but they are not ordered.
  - Examples: gender; blood type, name of fruits and their production regions etc.




Data
Qualitative/ Categorial
Quantitative
Discrete
Continuous
Levels of measurement
Nominal (e.g., gender, religion)
Ordinal (Can be ordered, e.g., small/medium/large)
Interval (No natural zero, e.g., temperature in Celsius)
Ratio (Includes natural zero, e.g., temperature in Kelvins)
Binary (e.g., True/False)
Non-Binary
Binary (e.g., undergrad/postgrad)

Data Types

- Nominal data: Nominal data attributes are also called categorical or qualitative data attributes. With some exceptions, the categories are distinct but have no explicit ranking or order. Such attributes often appear in natural settings as the attribute label or name form.
- Ordinal data: For data of ordinal scale, the exact numerical value has no significance over the ranking order that it carries. In other words, the categories have an explicit order, but the difference between them is not necessarily uniform or measurable. Such data is often seen in social and medical sciences when certain survey or health conditions are coded in importance level or strength level.
- Interval and ratio data: Both interval attributes and ratio attributes of data are represented by numerics. However, they are different in terms of whether the natural (absolute) zero-point is included. The ratio scale has an absolute zero-point. For example, age, height and weight never fall below zero. For the interval scale, the zero-point is arbitrary where negative values is permitted.

Nominal data

Examples

- With no intrinsic order: gender (male or female), race, blood type, place, identification number, name of fishes, name of fruits, and etc. These data can be represented in summary form, either in the form of percentage or in terms of frequency of occurrence. However no *mean* value regarding the labels can be defined.
- With intrinsic order: color where the underlying electromagnetic wavelength can be measured, age where the measurement is time based, and educational level where an intrinsic order can be established.

Nominal data

Numeric Conversion

- Arbitrary assignment: Assign arbitrary numbers to represent these attributes.
  - For example, one can assign a '1' value for male and a '2' value for female for the case of gender attribute. However, the label which is assigned a higher value may have a greater influence than the one with a lower value when extremely large values and extremely small values are involved along the computational process.
  - Moreover, the intrinsic relationship among the nominal attributes may be lost when the numerical assignment is arbitrary. For example, when representing the directional attributes 'north', 'east', 'south' and 'west' using '1', '2', '3' and '4' respectively, the neighborhood relationship among the attributes is lost.

Nominal data

Numeric Conversion

- Binary coding: Depending on the number of representing bits used, the binary coding offers a rich sources to represent nominal attributes in terms of binary numbers.
  - *binary-coded decimal*, *n-ary grey codes*, and *one-hot encoding*.
  - Sophisticated coding schemes take into account the probability distribution of each attributes during conversion.
  - After conversion, the binary feature vectors can be compared based on various distance measures such as Hamming distance and Spearman distance.

Ordinal data

Examples

- In the patient queue management system, patients who are either in pain or in frail health are often given a higher priority than those without severe symptoms.
- In a typical survey, several levels of response preferences are often assigned to represent the order of preference in the observation.

Ordinal data

Numeric Conversion

- Similar to the nominal data, the percentage or the frequency of occurrence can be quantified.
- Unlike the nominal case, the *mean*, *median* and *mode* values can be calculated when the ranks are assigned a numerical value.
- However, care should be taken for using the *mean* value since there is no knowledge regarding the distance among the ranks.

Ordinal data

Numeric Conversion

- The rating and ranking orders can be transformed into quantitative variables through normalization. Suppose the ranks are given by $r = 1, \ldots, R$. Then, the ranks can be normalized into standardized distance values (d) which fall within $[0, 1]$ using

$$d = \frac{r - 1}{R - 1}. \tag{1}$$

- For comparing the similarity or dissimilarity between two ranks, several distances such as Spearman distance, Hamming distance, and Chebyshev distance can be used.

Ordinal data

Numeric Conversion

- The Spearman distance is proportional to the square of Euclidean distance between two rank vectors. For example, given that the preferences of person x and person y on fruits given by [1:apple, 2:orange, 3: kiwi] are respectively [2,3,1] and [3,2,1]. Then, the Spearman distance of preference between x and y is proportional to

$$\begin{aligned} d(x, y) &= (x_1 - y_1)^2 + (x_2 - y_2)^2 + (x_3 - y_3)^2 \\ &= (2-3)^2 + (3-2)^2 + (1-1)^2 = 2. \end{aligned} \tag{2}$$

- The Hamming distance between two strings of ranks of equal length is the number of positions at which the corresponding ranks are different. For example, the Hamming distance between [2,3,1] and [3,2,1] is 2 since the first two ranks values are different.
- The Chebyshev distance is also called *maximum value distance* which measures the maximum absolute difference between two ordinal vectors.
- Other distances for ordinal variables include Kendall distance, Cayley distance, and Ulam distance.

Interval and ratio data

The ratio scale has an absolute zero-point. For example, age, height and weight never fall below zero. For the interval scale, the zero-point is arbitrary where negative values is permitted.

Example

- The temperature degree in Celsius is considered an interval attribute, since the "zero-degree" does not correspond to "no temperature" where we cannot say that 60 degrees Celsius is twice as hot as 30 degrees Celsius.
- However, for the Kelvin scale, the zero point represents the complete absence of molecular motion where we can say that 60 Kelvins is twice as hot as 30 Kelvins. Hence, the Kelvin scale is a ratio attribute.
- Although a distance measure between two points can be made for both ratio and interval attributes, scaling of these two attributes produces different interpretations.
- Hence, for these two reasons (absolute zero value, and the interpretation of scales), special attention needs to be paid for the effect of scaling during data normalization.

Distance metrics

Definition

A *metric* or a *distance function* refers to a function that defines a distance between each pair of elements of a set. Mathematically, the distance mapping function can be written as

$$d(x, y) : \mathcal{X} \times \mathcal{X} \mapsto [0, \infty), \tag{3}$$

where $[0, \infty)$ is the set of nonnegative real numbers. The following conditions are satisfied

$$\begin{aligned}
\text{Non-negativity axiom:} \quad & d(x, y) \geqslant 0, && (4)\\
\text{Identity of indiscernibles:} \quad & d(x, y) = 0 \quad \Leftrightarrow \quad x_1 = x_2, && (5)\\
\text{Symmetry:} \quad & d(x, y) = d(y, x), && (6)\\
\text{Triangle inequality:} \quad & d(x, z) \leqslant d(x, y) + d(y, z), && (7)
\end{aligned}$$

for all $x, y, z \in \mathcal{X}$. A set with a metric is called a metric space.

Distance metrics

Fig. 3 illustrates the three distances between each pair of points x, y, z.

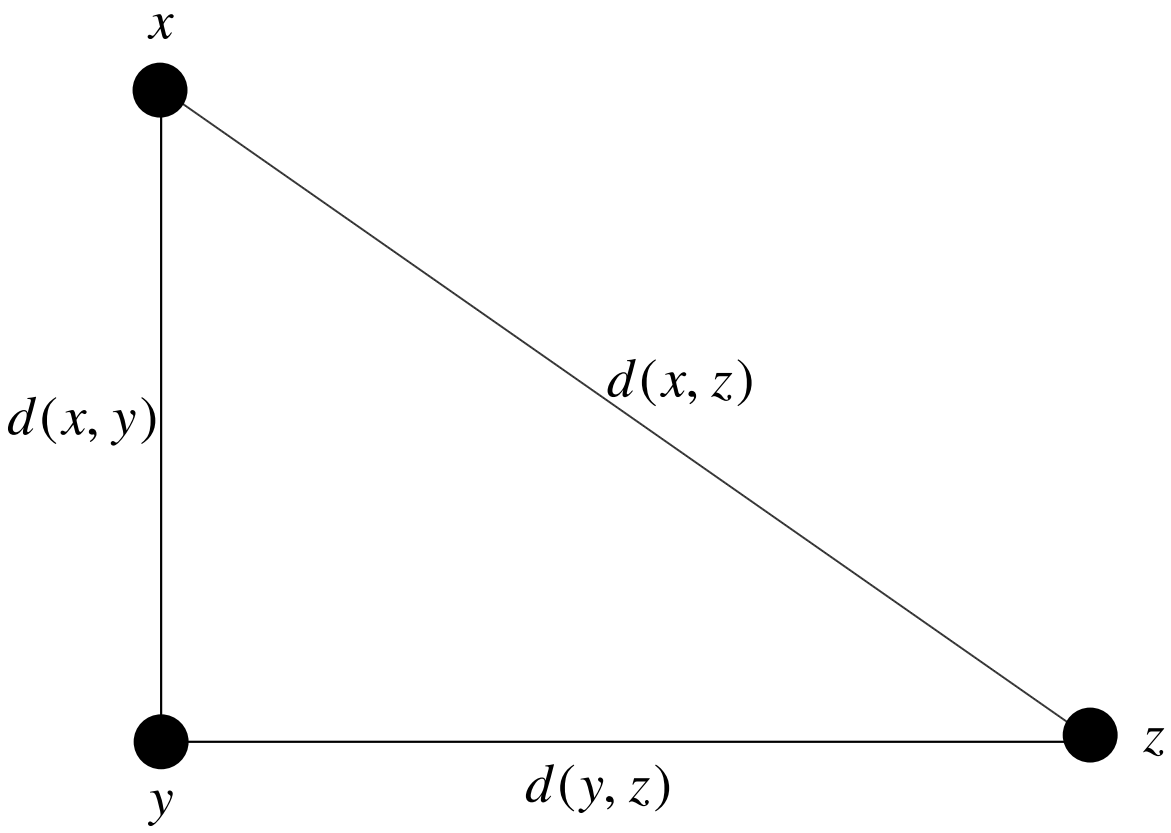


Figure 3: Three points form a triangle: $d(x, z) \leqslant d(x, y) + d(y, z)$.

Distance metrics

- The Hamming distance is a distance metric since it fulfills all the four metric conditions.
- Common examples of distance metrics include the Euclidean metric and the Manhattan or taxicab metric.
  - The Euclidean metric is also called the 2-norm or L_2-norm distance. It is similar to the geometrical distance between two points in space.
  - The taxicab metric is also call the 1-norm or L_1-norm distance.
- A generalization of both the Euclidean distance and the taxicab distance is called the Minkowski distance or p-norm metric.

Distance metrics

1-norm distance: $$\sum_{i=1}^{d} |x_i - y_i|$$

2-norm distance: $$\left(\sum_{i=1}^{d} |x_i - y_i|^2\right)^{1/2}$$

p-norm distance: $$\left(\sum_{i=1}^{d} |x_i - y_i|^p\right)^{1/p}$$

infinity-norm distance: $$\lim_{p\to\infty}\left(\sum_{i=1}^{d} |x_i - y_i|^p\right)^{1/p} = \max\left(|x_1 - y_1|, |x_2 - y_2|, \ldots, |x_d - y_d|\right).$$

The Minkowski distance is a metric for $p \geqslant 1$ since it satisfy all the four conditions listed in above. However, the triangle inequality is not satisfied for $p < 1$. Hence, the Minkowski distance is not a metric for $p < 1$.

Distance metrics

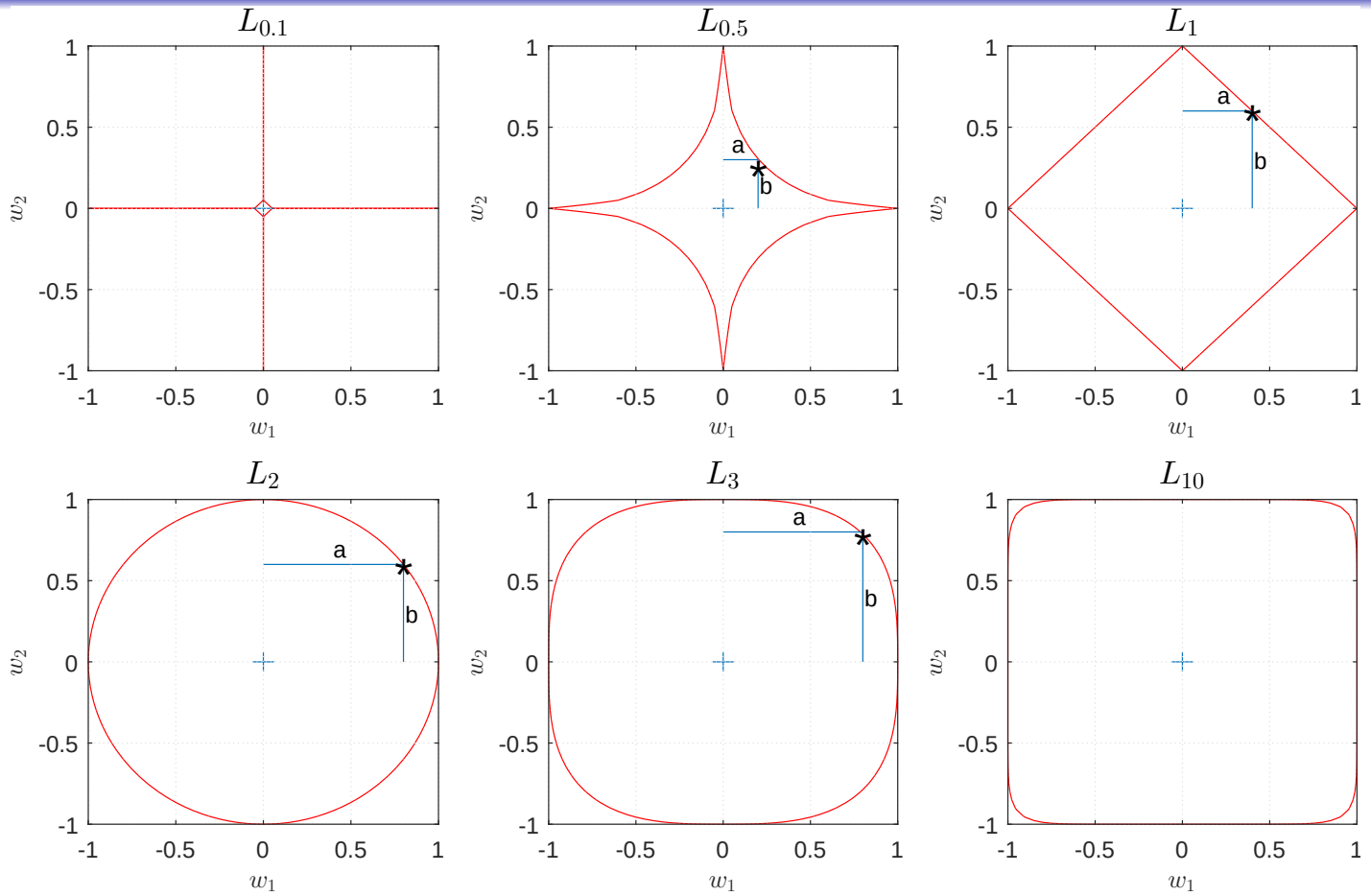


Figure 4: Minkowski distance (except for $L_{0.1}$ and $L_{0.5}$ which violate the triangle inequality, all other distances are distance metrics) on a two-dimensional space $\boldsymbol{w}$. The red solid contour indicates one unit distance from the origin. For example, in the L_1 plot, the distance between '+' and '$*$' is $d(+,*) = a + b = 1$. Other examples include $L_{0.5}$: $d(+,*) = (a^{0.5} + b^{0.5})^{1/0.5} = 1$; L_2: $d(+,*) = \sqrt{a^2 + b^2} = 1$; and L_3: $d(+,*) = (a^3 + b^3)^{1/3} = 1$.

Preprocessing

The collected data can come in any of the above forms namely, nominal, ordinal, interval and ratio types.
Moreover, multiple data sources can be collected in parallel and over a period of time. In other words, the data can come in large spatial dimensions and over time.
In order to achieve consistency in data representation, the following techniques are often adopted for data preprocessing.

Cleaning or cleansing

Data cleaning or cleansing is the process of detecting, correcting and removal of corrupted, incomplete, erroneous or inaccurate data samples from a collection.

- Completeness: This is the degree to which all required measurements are available. For some applications, the experiments can be repeated to collect more measurements. For others, those data with incomplete measurement are frequently removed.
- Consistency: This is the degree to which the measurements are reproducible under similar conditions. For stationary data, the measurements should not drift. For non-stationary data, the drift can be modelled.

Cleaning or cleansing

Data cleaning or cleansing is the process of detecting, correcting and removal of corrupted, incomplete, erroneous or inaccurate data samples from a collection.

- Uniformity: This is the degree to which the measurements are made using the same unit of scale and representation. For example, in one system the weight measurement could be in kilogram but in another system the weight measurement could be in pounds. If these data are to be put together, they have to be represented under the same scale.
- Validity: This is the degree to which the measurements conforms to well-established constraints. For example, for data represented in percentage, the total sum of such measurements must be equal to 100 percent.

Alignment

In image processing, a frequent preprocessing step is to extract relevant portion for subsequent processing steps.

For example, in face recognition, the face image excluding the hairs and wearing accessories is often extracted from the background image for comparison. Here, the alignment of face is often obtained based on the eyes, nose and mouth locations where only regions around these regions are used for comparison.

Normalization

- Min-max: When the bounds or range of each independent dimension of data is known, a common normalization technique is *min-max*. Alternatively, an estimation of the minimum and maximum values for a set of measurements can be made if the bounds are unknown. Given a set of raw measurements $\{x_i^{raw}\} \in [x^{min}, x^{max}]$, $i = 1, 2, ..., M$. Then the normalized measurements can be computed using

$$x_i = \frac{x_i^{raw} - x^{min}}{x^{max} - x^{min}}, \quad i = 1, 2, ..., M. \tag{8}$$

Normalization

- Standardization: When the population of measurements of each independent dimension of data is normally distributed where the parameters are known, the *standard score* or z-score is a popular choice.

$$x_i = \frac{x_i^{raw} - E[X]}{\sigma(X)}, \quad i = 1, 2, \ldots, M. \tag{9}$$

The exact values of $E[X]$ and $\sigma(X)$ may not be known in practice. Here, an estimation of the parameters can be made using a set of collected measurements:

$$\text{Sample mean} : \hat{\mu} = \frac{1}{M}\sum_{i=1}^{M} x_i, \tag{10}$$

$$\text{Sample variance} : \hat{\sigma} = \frac{1}{M}\sum_{i=1}^{M} (x_i - \hat{\mu})^2. \tag{11}$$

This normalization process is also called *standardization* in statistics.

Normalization

- Median Absolute Deviation: Instead of using the mean value as the reference point in standard deviation, the *median value* is adopted as reference point in the *median absolute deviation*:

$$\text{MAD} = \text{median}(|x_i - \text{median}(X)|), \tag{12}$$

where the normalization takes the form of

$$x_i = \frac{x^{raw} - \text{median}(X)}{\text{MAD}}. \tag{13}$$

Other Transformations

Apart from the above normalization schemes, other types of data transformation may be adopted to translate or stretch the data.
A simple transformation could be by means of translating the data towards a certain desired range of operation.
Operations such as multiplying, adding or subtracting a certain value uniformly to or from the measurement to get the data falling within a certain range provides a viable option to transform the data linearly.

Other Transformations

For measurements which spreads in exponential scale, taking the *logarithm* on such measurements (i.e., $\log(x_i)$) would be helpful to scale down the masking impact of large data values over small data values. Conversely, an exponential function (e.g., $\exp(x_i)$) can stretch the data when the measurements display a combination of flat and steep changing regions.

Some other nonlinear stretching include *sigmoid* and hyperbolic tanh functions:

$$x_i = \frac{1}{1 + e^{-h(x_i^{raw})}}, \tag{14}$$

$$x_i = \tanh(h(x_i^{raw})), \tag{15}$$

where the function $h(\cdot)$ can take any of the above normalization forms (such as (9)).

Thank you!